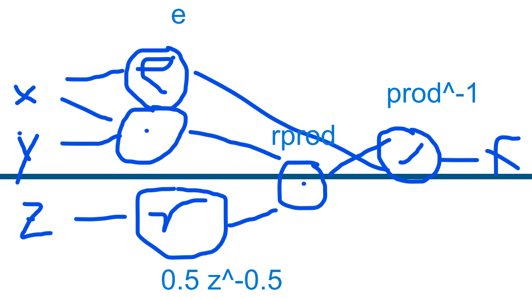
Exercise Sheet 3

Fundamentals



Automatic Differentiation

Exercise 3.1: Computational Graphs *optional*

Draw the computational graph for the function:

f over x, f over y and z over x are calculatable

$$f(x, y, z) = \frac{(x \cdot y)\sqrt{z}}{\exp(x)} \tag{1}$$

Write down some of the derivatives that you could compute with this graph (you do not need to compute the derivatives, just discuss which ones could be calculated and why).

Exercise 3.2: Backpropagation *optional*

Explain the purpose of the backpropagation algorithm.

applying derivatives from known losses to partially involved multipliers given the product rule -> distributing weights updates given their influence on the computation and loss

Exercise 3.3: Gradient Descent *optional*

Explain the idea behind the gradient descent algorithm. Write down the update formula. What would happen if the sign was inverted?

with gradient descent we try to gradually minimize the loss with steps in the direction of the inversed local gradient that gives us a local approximation. with the sign inverted we would follow the gradient to the highest loss and maximum missclassification

Practical: Training the CARLA Baseline Models

In this exercise you train the three models that will be the subject of your safety case. Use the provided CARLA dataset.

The dataset contains forward-facing camera images collected in CARLA under *clear weather* and *daytime* conditions, each annotated with three binary labels: *traffic light present*, *pedestrian present*, *vehicle present*.

You will train **three separate binary classifiers** – one for each detection task:

- A **pedestrian detector**
- A **traffic light detector**
- A **vehicle detector**

Each model takes a camera image as input and outputs a single binary prediction, trained with cross-entropy (CE) loss.

Exercise 3.4: Dataset Exploration

Load the dataset and explore it.

1. How many images are in the training and test splits? 7200 training pictures 3600 test
2. What is the class distribution for each label? Are the classes balanced?
3. Display a few example images for each label combination. What patterns do you notice? pattern: corners none, trafic lights at intersections, pedestrians infrequent, intersections mostly all

2 test: Total Frames Analyzed: 3600

Label	Total Activations	Activation %
has_traffic_light	2584	71.78%
has_pedestrian	706	19.61%
has_vehicle	2700	75.00%

2 train: Total Frames Analyzed: 7200

Label	Total Activations	Activation %
has_traffic_light	5276	73.28%
has_pedestrian	1718	23.86%
has_vehicle	5458	75.81%

Task	Accuracy	Precision	Recall	F1-Score
has_pedestrian	0.7889	0.4375	0.0467	0.0843
has_traffic_light	0.8972	0.8949	0.9778	0.9345
has_vehicle	0.8306	0.8817	0.9037	0.8926

Exercise 3.5: Train Three Binary Classifiers

Train three separate binary classifiers on the training split – one for each detection task (pedestrian, traffic light, vehicle). Use Python and preferably `PyTorch`. You may use a pre-trained backbone (e.g., ResNet-18) and fine-tune it for binary classification. Commit your code to your `git` repository.

1. Describe your model architecture and training setup (optimizer, learning rate, loss function). AdamW (weight decay), LR=0.0003 (reduce on plateu), cross entropyloss,
2. Train each model and plot the training and validation loss curves. Do the models converge? no, not really. just slowly
3. Why might training three separate models be preferable to a single multi-label classifier from a safety perspective? individuel components can be developed seperately. if one model is sufficient it might be used earlier than others. it is also more transparent in its logics for explainability. the frequency of predictions necessary might also change: OOD 1s, pedestr 1ms

Exercise 3.6: Evaluation

Evaluate each of your three trained models on the test split.

1. Report accuracy, precision, recall, and F1-score for each model.
2. Which model performs worst? Hypothesize why. predestrian is worst, the recall is realy bad, the others are ok
3. From a safety perspective, which metric matters most for each model – precision or recall? Explain. recall, because in an hazardoes environment, the safety system should break. however for airbags precision is more important as activating those is a hazard for the driver

Save all three trained models. You will use them in all subsequent exercise sheets.

ODD Gap Analysis

Exercise 3.7: Comparing the ODD to the Training Data

In Exercise Sheet 2 you defined an Operational Design Domain (ODD) based on the system description – before seeing any data. Now that you have explored the dataset and trained your models, revisit that ODD.

1. Look at the data you trained on. What conditions does it cover? List the environmental conditions represented in the training set (weather, lighting, scene types, etc.).
2. Compare this to the ODD you defined in Exercise Sheet 2. Which ODD dimensions are *not* covered by the training data?
3. What are the implications of this gap for your models' safety? For which ODD conditions can you *not* make any claims about model performance?
4. Note down the gaps you identified (if any). You will revisit them in Sheets 4 and 5 when you formally measure ODD coverage and refine the ODD.

sunny, good lighting, urban

2.) no fog, rain, night, tunnel, rural, highway,mountains, waterways, railways

3.)all of the conditions above

4.)

fog and rain and diffent urban environment in test set

— has_pedestrian

	precision	recall	f1-score	support
absent (0)	0.80	0.98	0.88	570
present (1)	0.44	0.05	0.08	150
accuracy			0.79	720
macro avg	0.62	0.52	0.48	720
weighted avg	0.72	0.79	0.71	720

— has_traffic_light

	precision	recall	f1-score	support
absent (0)	0.91	0.66	0.76	180
present (1)	0.89	0.98	0.93	540
accuracy			0.90	720
macro avg	0.90	0.82	0.85	720
weighted avg	0.90	0.90	0.89	720

— has_vehicle

	precision	recall	f1-score	support
absent (0)	0.63	0.57	0.60	159
present (1)	0.88	0.90	0.89	561
accuracy			0.83	720
macro avg	0.75	0.74	0.75	720
weighted avg	0.83	0.83	0.83	720